Practice Set 7 Paper 3 – Subspecialities of Urology

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Practice paper compiled from the question bank.

Q1. Mention in brief, urological complications in OBS-GYN. Describe in detail treatment of ureterovaginal fistula.

Type: LAQ (25 marks) (OBGY complications also answered in Practice Set 3 Paper 3, Q1; ureterovaginal fistula in 2015-Summer Paper 3, Q3e and 2017-Summer Paper 3, Q3b)

Assumed meaning: OBS-GYN = obstetric and gynaecological practice and surgery.

Part A: Urological complications in OBS-GYN

- The bladder, distal ureter and urethra lie against the cervix, uterine vessels and anterior vaginal wall. Risk rises with distorted anatomy: fibroids, endometriosis, cancer, previous caesarean, radiation, heavy bleeding.

Setting	Complications
Obstetric	**Obstructed labour** (fetal head impacted for days): pressure necrosis, so large VVF (vesicovaginal fistula), urethral loss, ureteric involvement; over 90% of VVF in the developing world. **Caesarean**: bladder injury (about 0.3%), ureteric injury, **vesicouterine fistula** (Youssef syndrome: cyclical haematuria, amenorrhoea, continence preserved). Post-partum or post-epidural **retention**
Gynaecological surgery	**Bladder injury** (commonest organ; hysterectomy about 0.3–0.75%). **Ureteric injury** (hysterectomy 0.2–2.4%; over 90% distal third; ligation commonest). **Fistulae**: VVF (over 75% follow hysterectomy in the developed world), UVF (ureterovaginal fistula), urethrovaginal fistula
Radical hysterectomy	Pelvic plexus injury: retention, poor compliance; manage with CIC (clean intermittent catheterisation)
Prolapse and continence surgery	Ureteric kinking at apical suspension; mesh or sling exposure, perforation, outlet obstruction, pain
Radiotherapy	Radiation cystitis, VVF, ureteric stricture

- **Prevention:** know the four ureteric danger sites; cystoscopy after hysterectomy where indicated; prophylactic stents only selectively (they aid recognition but do not prevent injury).

Part B: Ureterovaginal fistula (UVF)

Aetiology and presentation (brief)

- About **three-quarters follow an uneventful abdominal hysterectomy**: an unrecognised distal ureteric injury (ligation, crush, thermal, devascularisation) necroses and drains into the vaginal cuff. Diagnosis is delayed in 18–88%; **delayed detection raises fistula risk about fourfold**.
- **Continuous leakage with normal voiding** (the other kidney fills the bladder), usually **within 4 weeks**, often with fever or flank pain.

Confirming the diagnosis before treatment

Test	Method and reading
Double dye test	Oral phenazopyridine 100–200 mg; after 1 hour, blue dye into the bladder; tampon; walk. **Orange = UVF**; blue = VVF; both = both
CTU (computed tomography urography), delayed phase	Hydronephrosis, extravasation; MR (magnetic resonance) urography if contrast is contraindicated
Cystoscopy + RGP (retrograde pyelogram)	Abrupt cut-off or contrast leak; stent at the same sitting

- **Exclude a coexisting VVF** (up to 25% of VVF have a UVF) and bilateral injury.

Treatment

Goals: dryness and preservation of renal function. Management has moved from nephrectomy, through reimplantation, to endourology.

1. Endourological (first line)

- **Retrograde double-J stent as soon as possible.** Stenting **within 2 weeks** of detection succeeds far more often than at 2–6 weeks, because the ureter heals over the stent before fibrosis.
- Wire will not pass: **PCN (percutaneous nephrostomy)** with antegrade stent, or rendezvous.
- Stent for **4–6 weeks**, then remove and re-image. Most stentable fistulae close; a stricture may follow.

2. Surgical repair (stent fails or impossible)

- **Timing:** no consensus. Early repair took less time and blood loss with equal success; a systematic review found equal success before and after 6 weeks. Delay only if tissue is still inflamed.
- **Ureteroneocystostomy** (distal third, so usually feasible):
  1. Transperitoneal or extraperitoneal approach; identify ureter above the fistula.
  2. Divide above the diseased segment; dissect as distally as possible for length; **preserve periureteric adventitia**, especially laterally where vessels enter.
  3. Mobilise the bladder; spatulate; tension-free mucosa-to-mucosa anastomosis over a double-J stent; refluxing or tunnelled.
  4. Drain; catheter 7–14 days.
- **Adding length:**

Technique	Extra reach
Psoas hitch (sutures parallel to psoas fibres, spare genitofemoral nerve)	Up to about 5 cm
Boari flap (base ≥4 cm, length:base ≤3:1)	10–15 cm
Downward nephropexy; ileal ureter	Long defects

- Open, **laparoscopic and robotic repair give equal results**; vaginal reimplantation reported. **Success ≥90%.**
- The vaginal opening is left to heal; no separate vaginal repair is needed.

```
Suspected UVF -> exclude VVF (dye test, cystoscopy) -> CTU / RGP
  -> stent passes? Yes -> double-J 4-6 wk -> re-image -> healed / persistent
                  No  -> PCN + antegrade stent -> fails
  -> ureteroneocystostomy + psoas hitch / Boari flap
```

Complications and follow-up

- Hydronephrosis, infection, ischaemic stricture.
- Ultrasound 1–2 weeks after stent removal; some add a renal scan at 6 weeks. Stricture usually appears within a year but **up to 30 months**, so follow beyond this.

Diagram to draw: ureter passing under the uterine artery; ureteroneocystostomy with psoas hitch.

Key points to remember:

- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
- UVF: leak with normal voiding; orange tampon on double dye test.
- Stent within 2 weeks; remove at 4–6 weeks.
- Reimplant with psoas hitch or Boari flap; success ≥90%; follow 30 months.

Q2. Discuss the developmental basis of extrophy epispadias complex. Briefly outline management

Type: LAQ (25 marks) (theories also answered in Practice Set 6 Paper 1, Q2; single-stage repair in Practice Set 3 Paper 3, Q3e)

Introduction

- **EEC (exstrophy–epispadias complex)** is one spectrum: glanular epispadias → **CBE (classic bladder exstrophy)**, over 50% → cloacal exstrophy.
- CBE about 1 in 46,000; male:female 2.3:1. Offspring of an affected parent: 1 in 70.

Developmental basis

1. Normal development

- The **cloacal membrane** (bilaminar ectoderm–endoderm) occupies the infraumbilical wall. **Mesenchyme grows in** between its layers and forms the lower abdominal muscles and pelvic bones.
- The **urorectal septum** divides the cloaca into bladder (anterior) and rectum (posterior), meets the membrane, which perforates into urogenital and anal openings. Paired genital tubercles fuse **cephalad** to the membrane.

2. Core defect

- The cloacal membrane is **not reinforced by mesodermal ingrowth**, so it ruptures. **Timing and extent of rupture decide the variant:** before septum descent gives cloacal exstrophy; after gives CBE; limited rupture gives epispadias.

3. Theories

Theory	Proposed defect	Consequence
Cloacal membrane overdevelopment (Marshall–Muecke)	Abnormally large membrane in week 4 blocks medial mesenchymal migration	Timing of rupture gives the variant
Lateral body-wall fold failure	Folds fail to meet in midline	Abdomen and pelvis → cloacal; pelvis alone → classic
Caudal genital hillocks	Tubercles fuse **below** the membrane	Dorsal urethral defect (epispadias)
Abnormal body-stalk insertion	Caudal insertion stops mesenchyme interposing	Unstable membrane disintegrates
Bony pelvis first	Pelvic ring primordium fails to rotate	Bladder herniates

- All converge on **absent infraumbilical midline mesoderm**.

4. Aetiology

- Multifactorial (inconsistent twin concordance). **22q11.2 duplication** commonest variant; **ISL1** susceptibility gene; p63 under-expressed in exstrophic urothelium.
- **IVF (in vitro fertilisation)** about 7.5-fold and early high-dose progesterone about tenfold raise risk; maternal smoking linked to cloacal exstrophy; folate to the mildest form.

5. Resulting anatomy (why each defect occurs)

- **Pelvis:** externally rotated (posterior 12°, anterior 18° each side), **anterior pubic segment 30% short**, **diastasis mean 4.8 cm**.
- **Male genitalia:** anterior corpora about 50% shorter; short, upturned penis with dorsal chordee.

- **Female:** bifid clitoris, short vagina, anterior stenotic introitus; later prolapse.
- **Bladder:** immature, not neuropathic (raised type III collagen falls after closure). Ureters enter with no obliquity, so **VUR (vesicoureteric reflux) in 100%** after closure.
- Anterior anus, wide levator hiatus, inguinal hernias, low umbilicus.

Outline of management

At birth

- Tie cord with silk; cover the plate with plastic wrap; saline irrigation. Counselling; deliver in an exstrophy centre. Prenatal diagnosis only about 25%: absent bladder filling, low umbilicus, wide pubic rami, small genitalia, enlarging lower abdominal mass.
- **Assess:** template size (**≥3 × 3 cm** suits newborn closure), polyps, diastasis, pelvic malleability.
- Small template, penoscrotal duplication, hypoplastic bladder: **delayed closure at 6–9 months**.

MSRE (modern staged repair of exstrophy)

- Aim of closure: convert exstrophy into incontinent epispadias, with enough outlet resistance for bladder growth but not enough to harm the kidneys.

Stage	Age	Procedure
1	Newborn	Bladder, posterior urethra and abdominal wall closure; ± osteotomy
2	6–12 months	Epispadias repair (**modified Cantwell-Ransley**)
3	5–7 years (capacity ≥100 mL)	Bladder neck reconstruction (**modified Young-Dees-Leadbetter**) + Cohen reimplantation

- **Osteotomy** (combined anterior innominate + vertical iliac, external fixator, Buck traction 4–6 weeks): if closure after 72 hours, diastasis >4 cm, poorly malleable pelvis, failed closure, or cloacal exstrophy. Under 72 hours with malleable pubes: modified Bryant traction for 4 weeks.
- Key to closure: **radical division of the urogenital diaphragm fibres** so the bladder sits deep in the pelvis.
- **Alternatives: CPRE (complete primary repair, Mitchell)** with penile disassembly; Kelly radical soft-tissue mobilisation; Erlangen total repair; Mainz ureterosigmoidostomy.

Outcomes

- Daytime dryness after MSRE about **70–74%**; CPRE + YDL about 56%. **Bladder capacity at neck repair is the main predictor**; each failed closure lowers it.
- Failures: augmentation, bladder neck closure with continent catheterisable stoma.

Cloacal exstrophy (OEIS: omphalocele, exstrophy, imperforate anus, spinal defects)

- Stabilise; neurosurgery (spinal defects 85–100%); separate bowel from hemibladders; colostomy; osteotomy always; gender congruent with karyotype; continence mostly by augmentation with CIC (clean intermittent catheterisation).

Also know: variants (superior vesical fissure, duplicate, pseudoexstrophy); adenocarcinoma risk (cytology, cystoscopy); fertility and prolapse in adults; female epispadias.

Diagram to draw: cloacal membrane with urorectal septum and failed mesodermal ingrowth.

Key points to remember:

- Failed mesodermal reinforcement of the cloacal membrane; rupture timing decides variant.
- Diastasis 4.8 cm; anterior pubis 30% short; VUR in 100%.
- MSRE: closure → epispadias repair 6–12 months → YDL at 5–7 years.
- Osteotomy if >72 h or >4 cm; capacity ≥100 mL predicts continence.

Q3a. Energy sources in lap uro.

Type: SAQ (10 marks) (also answered in 2015-Summer Paper 2, Q3d)

Principles

- All devices achieve haemostasis by **denaturing protein**, by heat (current, light) or mechanical vibration.
- **Electrosurgery** uses high-frequency alternating current (about 500 kHz) that does not stimulate nerve or muscle. **Current density** decides heating: the small tip burns, the large pad does not.

Mode	Waveform	Effect
Cut (50–80 W)	Continuous, low voltage	Cell water flash-boils; clean cut
Coagulation (30–50 W)	Interrupted (about 6% duty cycle), high voltage	Desiccation, deeper spread
Fulguration (spray)	Higher voltage, sparks	Surface charring of oozing beds
Blend	Cut with interruptions	Cut plus coagulation

Devices

Device	Principle	Lateral spread	Notes
Monopolar (hook, scissors)	Tip → patient → return pad	**5–7 mm**	Versatile; stray-current risk
Bipolar	Current only between jaws	1–6 mm	No pad; no capacitive coupling
Vessel sealer (LigaSure)	Bipolar + compression + impedance feedback; built-in blade	1–3 mm	Seals vessels **≤7 mm**; one application
Ultrasonic (Harmonic)	Blade vibrating at **25–55 kHz**	**≈1 mm**	No current: safe with pacemakers; jaws **>200 °C**, slow to cool, so do not touch bowel
Argon beam	Current through argon, non-contact	Superficial	Liver, spleen, raw surfaces; gas embolism

- **Also know:** lasers (KTP [potassium titanyl phosphate], holmium) rarely used; laparoscopic RFA (radiofrequency ablation, 60 °C) and cryoablation (below −40 °C, two freeze–thaw cycles); robotic HotShears, bipolar Maryland, Vessel Sealer.

Electrosurgical injuries

Mechanism	Prevention
Direct activation out of view	Activate only with tip in view
Direct coupling to another metal instrument	Whole metal tip visible
Capacitive coupling (metal cannula in plastic retainer)	Never mix metal and plastic ports
Insulation failure	Inspect; **AEM (active electrode monitoring)**
Pad burns; surgical fire	Pad on muscle, away from implants; REM (return electrode monitoring)

- Bowel injuries present late (days) as peritonitis.

Diagram to draw: monopolar vs bipolar circuits.

> **Key points to remember:**
>
> - Spread: monopolar 5–7 > bipolar 1–6 > sealer 1–3 > ultrasonic ≈1 mm.
> - LigaSure seals ≤7 mm; ultrasonic blade hottest after use.
> - Capacitive coupling: avoid hybrid trocars; AEM detects it.

Q3b. Describe in details anatomy of prostate with special reference to newer concept in surgical management of localized CA prostate

Type: SAQ (10 marks) (also answered in Practice Set 2 Paper 4, Q1)

Anatomy

- About 20 g; base at bladder neck, apex at urogenital diaphragm. **No true capsule**.

Zone (McNeal)	Glands	Relevance
Peripheral	70%	**70% of cancers**; posterolateral extension along nerves
Central	25%	Around ejaculatory ducts
Transition	5%	BPH (benign prostatic hyperplasia); anterior cancers
AFMS (anterior fibromuscular stroma)	None	Anterior margin

- **Fascial planes:** intrafascial (most nerves, more margins), **interfascial (standard)**, extrafascial (widest margin). Denonvilliers fascia excised if posterior extension suspected.
- **NVB (neurovascular bundle):** cavernous nerves posterolateral, **5 and 7 o'clock** at the apex; diffuse "hammock" distribution. Injured by heat and traction.
- **Accessory pudendal arteries:** preserve. **DVC (dorsal venous complex)** tamponaded by pneumoperitoneum.
- **Continence:** striated sphincter at apex; anterior support (puboprostatic ligaments, detrusor apron).
- **Lymphatics:** obturator, external and internal iliac, presacral nodes.

Newer surgical concepts

1. **RARP (robot-assisted radical prostatectomy):** Yaxley 2016 RCT (randomised controlled trial): function equal to open, less blood loss.
2. **Risk-graded nerve sparing** using mpMRI (multiparametric MRI) and biopsy, side by side.
3. **NeuroSAFE:** frozen section of posterolateral margin guides nerve sparing.
4. **Retzius-sparing** (Bocciardi 2010), through the pouch of Douglas: earlier continence, more margins anteriorly; **Hood technique** preserves anterior support.
5. **Apical dissection:** maximal urethral length, cold division (commonest margin site); posterior (Rocco) reconstruction.
6. **Extended PLND (pelvic lymph node dissection)** when nodal risk >5% (EAU [European Association of Urology]): captures about 63% of drainage; staging, not therapeutic.
7. **Single-port robot**, extraperitoneal; **focal therapy** of the index lesion (HIFU [high-intensity focused ultrasound], cryotherapy).

Diagram to draw: axial prostate with three fascial planes and NVB at 5 and 7 o'clock.

> **Key points to remember:**
> - Peripheral zone: 70% of cancers; no true capsule.
> - Interfascial plane standard; apex commonest margin site.
> - Retzius-sparing: early continence, more margins.

Q3c. APOGEE

Type: SAQ (10 marks)

Assumed meaning: the Apogee transvaginal mesh kit (American Medical Systems) for posterior vaginal wall and vault prolapse. It is not described in the chapter notes; facts below are general.

Definition

- A **prepackaged polypropylene mesh kit** placed transvaginally to support the **posterior compartment and vaginal apex** (DeLancey levels I and II). Anterior companion: **Perigee** (transobturator).
- Mesh: **type I macroporous monofilament polypropylene** (Amid classification), the only type that integrates into tissue.

Indications (historical)

- Symptomatic posterior and vault prolapse, mostly **recurrent** or with poor native tissue, avoiding abdominal surgery.

Technique (principle)

1. Midline posterior vaginal incision; dissect the rectovaginal space to the ischial spines.
2. Two small perianal incisions lateral and posterior to the anus.
3. Curved trocars pass through the **ischiorectal fossa**, piercing the iliococcygeus near the ischial spine, and deliver the mesh arms.
4. Mesh body fixed to the apex and laid over the rectovaginal septum **tension-free**; epithelium closed without trimming.

Complications

- **Mesh exposure** (IUGA/ICS [International Urogynecological Association/International Continence Society] term; "erosion" deprecated), extrusion, **perforation** into rectum or bladder.
- **Dyspareunia**, buttock and pelvic pain, mesh contraction, defecatory dysfunction, ischiorectal haematoma or abscess, rectal injury.
- **Management:** small exposure → vaginal oestrogen, then partial excision; pain or perforation → excision of mesh and arms.

Regulatory status

- **FDA (US Food and Drug Administration) 2011:** serious complications of transvaginal POP (pelvic organ prolapse) mesh "not rare", with no clear benefit over native repair. Reclassified high-risk in 2016.
- **2019:** FDA ordered remaining manufacturers to **stop selling transvaginal POP mesh**; Apogee-type kits are off the market. Synthetic mesh improves anatomy but complicates more (Cochrane 2016).
- This does **not** apply to mid-urethral slings or **sacrocolpopexy mesh**.

Current alternatives

- Native posterior colporrhaphy or site-specific repair; uterosacral suspension or **sacrospinous fixation**; **sacrocolpopexy** (abdominal, laparoscopic, robotic) for apical prolapse; colpocleisis in the frail.

Diagram to draw: posterior mesh with two arms through the ischiorectal fossae.

Key points to remember:

- Apogee: transvaginal posterior/apical polypropylene mesh kit.
- Exposure, dyspareunia and pain drove withdrawal.
- FDA 2019 stopped transvaginal POP mesh sales.

Q3d. Newer techniques of stone analysis

Type: SAQ (10 marks) (also answered in 2024-Winter Paper 4, Q3a)

Why analyse stones

- Analyse every first stone (AUA [American Urological Association], EAU [European Association of Urology]); repeat on recurrence despite treatment or early recurrence after clearance.
- Composition points to the cause: **uric acid** metabolic, **cystine** genetic, **struvite** infection.
- Composition can change over time (e.g. calcium oxalate converting to brushite), so re-analyse recurrent stones.

Limits of wet chemical analysis

- Detects ions, not **crystal phases**: cannot separate COM (calcium oxalate monohydrate) from COD (dihydrate), or apatite from brushite. High error rates; misses rare stones; needs a large sample.

Newer methods

Method	Principle	Comment
FTIR (Fourier-transform infrared) spectroscopy	Each compound absorbs a characteristic infrared spectrum	Commonest reference method; small sample; detects drugs and proteins
XRD (X-ray diffraction)	Crystal lattice gives a unique diffraction pattern	Gold standard for crystals; misses amorphous material
Polarising microscopy	Birefringence	Cheap; observer-dependent
Morpho-constitutional analysis (Daudon)	Stereomicroscopy of surface and section + FTIR	Subtypes point to cause (e.g. Ic: primary hyperoxaluria)
SEM (scanning electron microscopy) ± EDX (energy-dispersive X-ray)	Surface structure, elements	Research
Micro-CT (micro-computed tomography)	3-D, non-destructive	Research
Raman spectroscopy	Scattered light spectrum	Research

In vivo prediction

- **DECT (dual-energy CT):** two tube energies (e.g. 80 and 140 kVp) separate **uric acid from non-uric acid** stones.
- **Hounsfield units:** >1000 HU predicts SWL (shock wave lithotripsy) failure.
- **Endoscopic recognition** of morphology (lost when stones are dusted).

Significance in management

- **Uric acid:** oral dissolution (potassium citrate).
- **Struvite:** complete clearance, culture-directed antibiotics.
- **Cystine:** hydration, alkalinisation, tiopronin.
- **COM, brushite, cystine:** SWL-resistant; prefer endoscopy.
- **Brushite:** avoid over-alkalinisation.
- **2,8-DHA (dihydroxyadenine):** APRT deficiency, allopurinol. **Drug stones** (indinavir, triamterene): stop the drug.

Diagram to draw: COM (dumb-bell) vs COD (envelope) crystals.

> **Key points to remember:**
>
> - FTIR and XRD are reference methods; wet chemistry obsolete.
> - DECT identifies uric acid stones in vivo.
> - Composition guides dissolution, SWL choice and prevention.

Q3e. Describe principles of tissue engineering and what its status in managemet of urological diseases is.

Type: SAQ (10 marks) (also answered in 2023-Winter Paper 1, Q1 and 2022-Summer Paper 4, Q2)

Definition

- **Tissue engineering** applies engineering and life sciences to build biological substitutes that restore or improve tissue function (Langer and Vacanti, 1993).
- **Rationale:** bowel in the urinary tract causes mucus, stones, acidosis and cancer; grafts are scarce.

Principles

```
Biopsy -> cells expanded -> seeded on scaffold (+ signals)
       -> bioreactor conditioning -> implant -> host vascularises
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Element	Details
Cells	Autologous urothelial and smooth muscle cells (no rejection); adult stem cells (bone marrow, adipose, urine-derived; mainly paracrine); embryonic and iPSCs (induced pluripotent stem cells): tumour risk
Scaffolds	Biocompatible, porous, degrade as tissue forms. Natural (collagen), **acellular matrices** (bladder matrix, SIS [small intestinal submucosa]), synthetic (PGA, PLGA [polyglycolic, poly(lactic-co-glycolic) acid])
Seeding	Urothelium inside, muscle outside. **Unseeded** matrix works only for short onlay defects; tubes need seeded scaffolds or they contract
Signals	Growth factors (e.g. VEGF [vascular endothelial growth factor]), matrix and mechanical cues
Bioreactor	Perfusion and cyclic stretch to mature the construct
Vascularity	Oxygen diffuses only **100–200 µm**: main barrier; omental wrap, angiogenic factors

Status in urology

Organ	Status
Bladder	**Atala 2006**: seeded constructs in 7 myelomeningocele patients improved compliance; phase II trial showed **no significant gain**. Experimental
Conduit	Neo-Urinary Conduit trial: **half developed stenosis**
Urethra	Acellular onlay for short strictures; **Raya-Rivera 2011**: seeded tubular urethras in **5 boys**, functional up to 6 years
Sphincter	Muscle-derived cell injection for SUI (stress urinary incontinence): investigational
Vagina	Engineered vaginas in MRKH (Mayer–Rokitansky–Küster–Hauser) syndrome
Penis, kidney, testis	Corporal grafts, kidney organoids, spermatogonial stem cell banking: experimental

- **Recent advances:** 3D bioprinting, organoids, vascularised composite bladder allograft.
- **Challenges:** vascularity, innervation, fibrosis, cost, regulation.

Diagram to draw: biopsy → culture → scaffold → bioreactor → implant.

> **Key points to remember:**
>
> - Cells, scaffold, signals, bioreactor, blood supply.
> - Diffusion limit 100–200 µm.
> - Still experimental except short urethral onlays.

Q3f. HOLEP

Type: SAQ (10 marks) (also answered in 2020-Summer Paper 4, Q3d)

Read as: holmium laser enucleation of the prostate for BPH (benign prostatic hyperplasia).

Principle

- **HoLEP (holmium laser enucleation of the prostate)** removes the whole adenoma along the **surgical capsule**: the endoscopic equivalent of open simple prostatectomy.
- **Ho:YAG (holmium:yttrium–aluminium–garnet), 2120 nm**: water absorbs it; pulsed; **0.4 mm penetration**; coagulates vessels. Saline irrigation, so no TUR (transurethral resection) syndrome.

Indications

- **Size-independent** (<30, 30–80, >80 mL), with or without median lobe.
- **Anticoagulated or high bleeding risk** (AUA [American Urological Association]); alternative to open prostatectomy for large glands.

Steps

- 60–150 W laser, 550 µm end-firing fibre, 24–28 Fr scope; **2.0 J × 40–50 Hz** for enucleation.

1. Bladder-neck incisions at 5 and 7 o'clock (median lobe) or one at 6 o'clock.
2. Perimontanal incision; **early apical release** protects the sphincter.
3. Develop the capsular plane to 12 o'clock; push lobes into the bladder (**en bloc, two- or three-lobe**).
4. Haemostasis with a defocused beam.
5. **Morcellation** with a full bladder; three-way catheter.

Outcomes

- Versus **TURP (transurethral resection of the prostate)**: **better IPSS (International Prostate Symptom Score) and Qmax (maximum flow rate)**, durable; less bleeding, shorter catheter time and stay; longer operating time.
- **Only endoscopic procedure superior to TURP**; retreatment rare (whole transition zone removed).

Complications

- **Retrograde ejaculation about 75%**; transient storage symptoms and early SUI (stress urinary incontinence).
- BNC (bladder neck contracture) 0–3.2%; stricture; capsular perforation; ureteric orifice injury; **morcellator bladder injury**.
- Erectile function unchanged.

Limitation

- **Steep learning curve** (about 50 cases).

Recent advances

- **Moses pulse modulation**; ThuLEP (thulium enucleation); median-lobe-only HoLEP (antegrade ejaculation about 90%); same-day discharge.

Diagram to draw: sagittal prostate with enucleation plane along the capsule.

Key points to remember:

- Ho:YAG 2120 nm, water chromophore, 0.4 mm.
- Size-independent; suits anticoagulated men.
- Only endoscopic method superior to TURP; retrograde ejaculation about 75%.

Rapid Revision Sheet – Practice Set 7 Paper 3

All the key points from this paper in one place. Read this in the last 30 minutes before the exam.

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- Bladder is the commonest organ injured; obstructed labour causes most fistulae worldwide.
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Q3a. Energy sources in lap uro. (SAQ)

- Spread: monopolar 5–7 > bipolar 1–6 > sealer 1–3 > ultrasonic ≈1 mm.
- LigaSure seals ≤7 mm; ultrasonic blade hottest after use.
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Q3b. Describe in details anatomy of prostate with special reference to newer concept in surgical management of localized CA prostate (SAQ)

- Peripheral zone: 70% of cancers; no true capsule.
- Interfascial plane standard; apex commonest margin site.
- Retzius-sparing: early continence, more margins.

Q3c. APOGEE (SAQ)

- Apogee: transvaginal posterior/apical polypropylene mesh kit.
- Exposure, dyspareunia and pain drove withdrawal.
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